

AtlasSAT-16U

Souss Water Observation CubeSat

Engineering Portfolio | Structural & Systems Design | Wassim Ait Oubahou
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ABOUT THIS PORTFOLIO

AtlasSAT-16U is an independent, end-to-end 16U CubeSat design project conducted to demonstrate mechanical and systems engineering competency across structural verification, CAD and manufacturing-ready drawings, and systems-level design. All analyses follow ECSS and Falcon 9 RPUG standards. No institutional affiliation.

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2	3A : Structural Assembly & Drawings	CAD assembly, configuration, manufacturing drawings
3	3B : Static Structural Analysis	FEA methodology, load cases, margins of safety

KEY RESULTS

- All six quasi-static load cases verified per ECSS-E-ST-32C & Falcon 9 RPUG r2.0
- Governing case QSL-04 (+Z, 5g): $MoS_{yield} = +7.40$, $MoS_{ult} = +6.88$
- First structural mode $f_1 = 503$ Hz, +403% above the 100 Hz minimum requirement

Tools	Use case
Solidworks	CAD Geometry Assembly
Solidworks Simulation	Static & modal FEA
Ansys Mechanical	Static & modal FEA
GMAT	Orbital propagation
VSC (Python)	Thermal transient simulation
MATLAB	Power budget modelling
Simulink	System architecture
LaTeX	Documentation

ABOUT

AtlasSAT-16U is an independent end-to-end CubeSat design project, conducted solely to demonstrate mechanical and systems engineering competency across structure, thermal, power, orbital mechanics and ADCS systems. All analyses follow ECSS and Falcon 9 RPUG standards. No institutional affiliation.

MISSION CONCEPT

AtlasSAT-16U-SWM is a 16U CubeSat designed to deliver multispectral VNIR imagery over the Souss-Massa basin in Morocco. The mission targets irrigated agriculture monitoring, reservoir extent mapping, and seasonal water body detection through NDVI, NDRE and ET-proxy indices. Data products are designed for water resource agencies and agricultural cooperatives operating in a water-stressed semi-arid environment.

OBJECTIVES

Document 3A is a portfolio overview of the AtlasSAT-16U primary structure, SolidWorks CAD assembly and representative 2D manufacturing drawings. It supports engineering demonstrations of structural design competency. Full drawings, native CAD files and a detailed BOM are held in companion report 3A-1.

CONFIGURATION

Parameter	Value
Form factor	16U (4 × 4 × 1U)
Deployer iface	P-POD / ISIPOD compliant
Rails	AI 7050-T73510 (×4)
Bulkheads	AI 6061-T6 (top, mid, bot)
Shear panels	AI 6061-T6 (×4)
Equipment plates	AI 6061-T6
Total mass	≈23 kg (FEA)
Struct. mass	7.90 kg (33%)
CoM (Y-axis)	235.7 mm
$I_{xx,yy,zz}$	0.199 / 0.641 / 0.661 kg·m ²
CAD tool	SolidWorks (ver. 2024)

SUBSYSTEM LAYOUT

Stacking order along Y axis (launch direction):

Zone	Subsystem
Upper	PLD optics (nadir-facing aperture)
Upper-mid	PLD electronics, OBC, mass memory
Centre	EPS/PCDU + Li-ion battery (CoM)
Lower-mid	ADCS: RW cluster, MTQ, IMU
Lower	COMMS: TT&C, X-band downlink

DESIGN PHILOSOPHY

- Stiffness-driven primary structure
- Clear load paths: deployer rails → bulkheads
- Conservative bonded contacts (preliminary)
- $f_1 = 503 \text{ Hz} \gg 100 \text{ Hz}$ RPUG minimum
- Lateral (Z) case governs: QSL-04 +Z 5g

CAD ASSEMBLY VIEWS

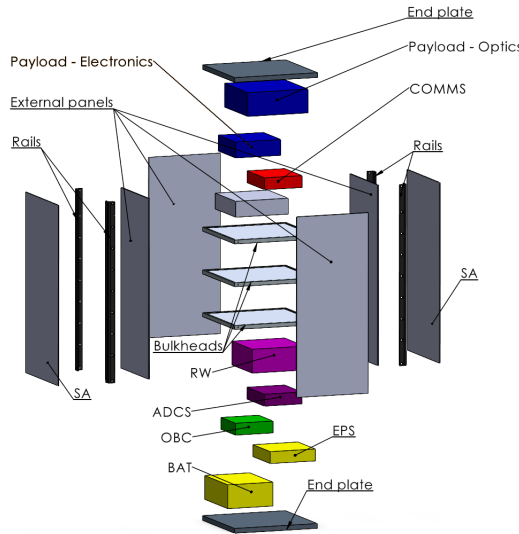


Fig. 1 — Exploded view. All structural elements and major subsystems labelled. Top-level file: *AtlasSAT16U_Assembly.SLDASH*.

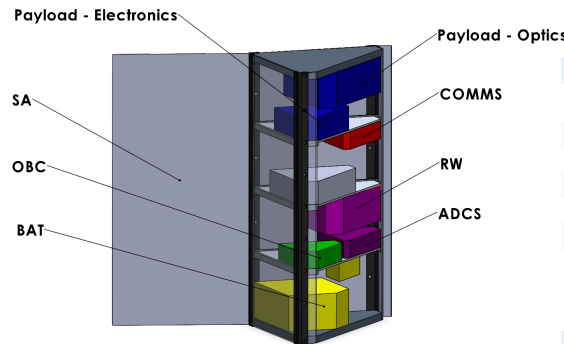


Fig. 2 — Section cut along Y (axial) axis showing internal stacking: PLD optics → OBC → EPS → ADCS → COMMS.

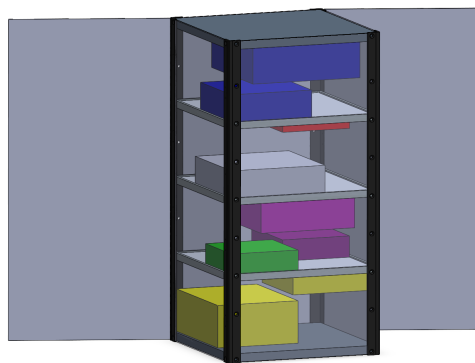


Fig. 3 — Full assembly isometric view.

ENGINEERING DRAWINGS

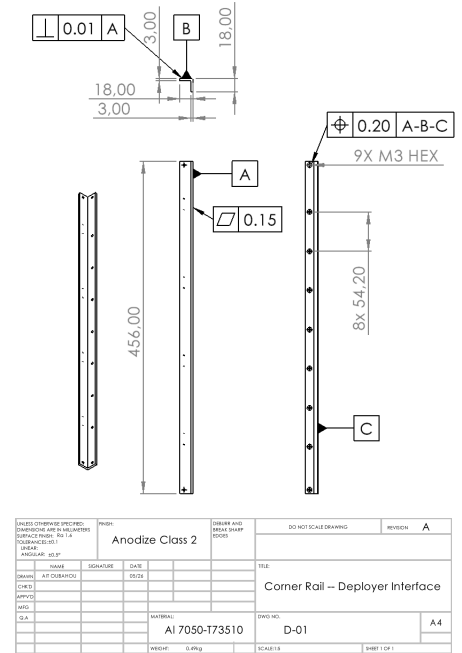


Fig. 4 — 2D drawings in SolidWorks: Corner rail D-01

Dwg Description

- D-01 Corner rail / deployer interface
- D-02 Top & bottom bulkhead
- D-03 Mid bulkhead (equipment deck)
- D-04 Side shear panels (×4)
- D-05 Payload mounting bracket
- D-06 Equipment plate (ADCS/EPS zone)

MASS BUDGET

Subsystem	Components	kg	%
Structure	Rails, panels, plates	7.90	33
PLD optics	VNIR imager, optics	4.50	18.8
PLD electronics	Payload electronics board	1.00	4.2
ADCS	Magnetorquers	0.80	3.3
RW	Reaction wheels ×3	2.00	8.3
Battery	Li-ion pack	3.00	12.5
EPS / PCDU	Solar arrays + distribution	1.50	6.25
OBC	Computer	0.60	2.5
COMMS	TT&C + downlink chain	0.80	3.3
Misc	Harness, brackets, misc	1.50	6.25
Total (FEA)		≈24	—

STRUCTURAL RESULTS

Check	Result
MoS _{yield} (QSL-04)	+7.40 > 1.25 req. DONE
1st eigenfreq. f_1	503 Hz $\gg$ 100 Hz DONE
Cold-case ΔT	17 K (246–263 K) DONE
Hot-case ΔT	96 K (260–347 K) WIP
Eclipse-case ΔT	/ WIP
Transient cold-case ΔT	/ WIP
Transient hot-case ΔT	/ WIP
Norms: ECSS-E-ST-32C · RPUG r2.0	

STRUCTURAL VERDICT — PASS

All 6-axis QSL cases satisfy ECSS-E-ST-32C.

$$MoS_{yield,min} = +7.40 > 1.25$$

$$MoS_{ult,min} = +6.88 > 1.50$$

Structure cleared for 3C and 3D.

ABOUT

AtlasSAT-16U is an independent end-to-end CubeSat design project demonstrating mechanical and systems engineering competency. Document 3B presents the static structural (quasi-static load) analysis of the primary structure, conducted in SolidWorks Simulation 2024. All loads and safety factors follow ECSS-E-ST-32C and the Falcon 9 RPUG r2.0.

OBJECTIVE

Demonstrate that the AtlasSAT-16U primary structure sustains launch quasi-static loads with positive margins of safety at yield and ultimate, while meeting the launcher's minimum frequency requirement. This one-pager summarises the analysis model, load cases, key stress and displacement results, and design-driven changes. Extended mesh convergence data and full stress reports are in companion report 3B-1.

ANALYSIS MODEL

Parameter	Value
FEA tool	SolidWorks Simulation 2024
Geometry	Full assembly: L-rails, bulkheads, panels, subsystem blocks
Masses	Subsystems as density-adjusted solid blocks
Element type	Solid (curvature-based, high quality)
Mesh — global	3 mm max; 1 mm min
Jacobian	16 points
Nodes / elem.	1.3 M / 0.87 M (fasteners excluded)
Material	Al 6061-T6 (all primary structure)
σ_y	276 MPa
σ_{ult}	310 MPa
Contacts	Bonded (global, conservative)

BOUNDARY CONDITIONS

Fixed Geometry applied to the four L-rail lower end faces (face $-Y$, 17×17 mm per rail), representing the ISIPOD deployer contact. All 6 DOFs constrained. Fasteners excluded from simulation. Inertial loads applied via Gravity (axial cases, 58.86 m/s^2 at 6 g) and Remote Load/Distributed at assembly CoM (lateral cases). Safety factors per ECSS-E-ST-32C Table 5-1.

DESIGN PHILOSOPHY

- Stiffness-driven primary structure
- Clear load paths: deployer rails $\rightarrow$ bulkheads
- Conservative bonded contacts (preliminary)
- Lateral (Z) case governs: QSL-04 +Z 5 g
- $FS_{yield,min} = 7.40 \gg 1.25$ requirement
- Structure cleared for modal and thermal phases

LOAD CASES (FALCON 9 RPUG r2.0)

ID	Axial (Y)	Lateral (X or Z)
QSL-01	+6 g	—
QSL-02	−3 g	—
QSL-03	—	+5 g (X)
QSL-04	—	+5 g (Z) — governing
QSL-05	+6 g	+2 g (X)
QSL-06	+6 g	+2 g (Z)

Safety factors: 1.25 (yield) and 1.50 (ultimate) per ECSS-E-ST-32C.

FEA FIGURES

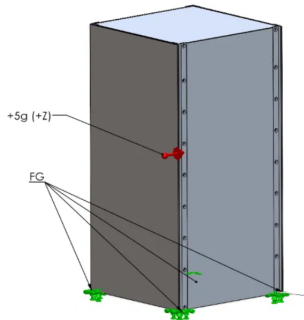


Fig. 1 — Boundary conditions and applied load, QSL-04 (+Z, 5g)

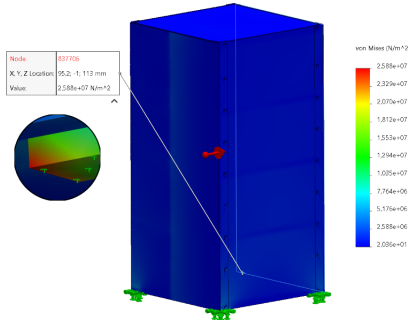


Fig. 2 — von Mises stress, QSL-04 (+Z, 5g, governing). Peak $\sigma = 25.88 \text{ MPa}$ at rail interface, node 837706 ($X = 95.2$, $Y = -1$, $Z = 113 \text{ mm}$); deformation exaggerated $\times 168$ for visualization.

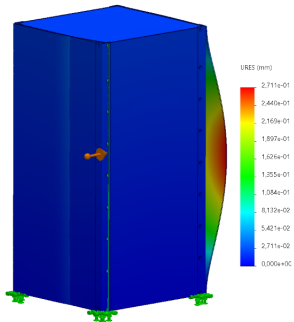


Fig. 3 — Total displacement (URES), QSL-04 (+Z, 5g). Max displacement = 0.271 mm; deformation exaggerated $\times 168$ for visualization.

MARGINS OF SAFETY SUMMARY

Case	σ_{max}	Allow.	MoS _{yield} / MoS _{ult}	Location
QSL-01	15.10	276	+13.61 / +12.69	Rail interface
QSL-02	7.55	276	+28.21 / +26.34	Upper bulkhead
QSL-03	23.42	276	+8.40 / +7.82	Shear panel
QSL-04	26.22	276	+7.40 / +6.88	Rail interface
QSL-05	18.84	276	+10.74 / +9.98	Rail interface
QSL-06	23.09	276	+8.51 / +7.93	Rail interface

All stresses in MPa. MoS = (Allowable / (SF $\times$ σ_{max})) − 1. Min. MoS > 0 required.

STRUCTURAL RESULTS

Check	Result	Status
FS_{yield} (QSL-04)	$7.40 > 1.25$ req.	DONE
FS_{ult} (QSL-04)	$6.88 > 1.50$ req.	DONE
f_1 (1st eigenfreq.)	$503 \text{ Hz} \gg 100 \text{ Hz}$	DONE
Max displacement (PLD)	0.271 mm	DONE
Mesh convergence	TBD	WIP
All 6 QSL runs complete	6/6	DONE
Norms: ECSS-E-ST-32C · RPUG r2.0		

DESIGN IMPACT

QSL-04 governs with $\sigma_{max} = 26.22 \text{ MPa}$ at the rail interface under lateral 5 g load. All margins positive. Structure is conservative — no design modifications required at this stage.

OPEN ITEMS (3B)

ID	Action	Pri.
OI-B1	Insert Figs. 1–3 SW Simulation screenshots	DONE
OI-B2	MoS table complete (6/6 cases)	DONE
OI-B3	Mesh convergence study (h-refine)	HIGH
OI-B4	Max displacement at PLD mount	DONE
OI-B6	All QSL runs completed	DONE

STRUCTURAL VERDICT — PASS

All 6-axis QSL cases satisfy ECSS-E-ST-32C.

$$FS_{yield,min} = 7.40 > 1.25 \mid FS_{ult,min} = 6.88 > 1.50$$

Structure cleared for 3C-1 modal and 3D-1 thermal.